Insurance campaign response: Policy & Payment

Pacharapol Vorravanpreecha (Pete) · 3 November 2026 · Data: dsc_test_case.csv

214,053 bank customers were offered a PA (personal accident) or life insurance policy over 12 campaign months. **Policy** = whether they bought (🟦 PA, 🟧 Life). **Payment** = what they paid from their bank account in the 30 days before the offer.

> Payment here is **pre-offer bank behaviour, not premium**. The dataset has no premium or policy-status column, so this page measures who buys, not revenue. All rates are % of customers offered.

How to read: 🟦 PA is blue and 🟧 life is orange throughout.

Customers offered	Policies sold	PA policies
214,053	**2,622**	**1,634**
1,940 empty duplicate rows removed	1.22% acceptance	0.76% rate · 62% of sales

Life policies	Median payment, 30 days	Made no payment
988	**฿373**	**21%**
0.46% rate · 38% of sales	before the offer	of customers, last 30 days

PA is an active-account product

83% of PA buyers had money moving in or out of their account in the 30 days before the offer. PA rate 1.27% active vs 0.26% dormant.

Best PA prospects: new and active

Active accounts opened 1–12 months ago buy PA at **2.98%**, 3.9× the PA average (3.8% for men).

Life is a young-earner product

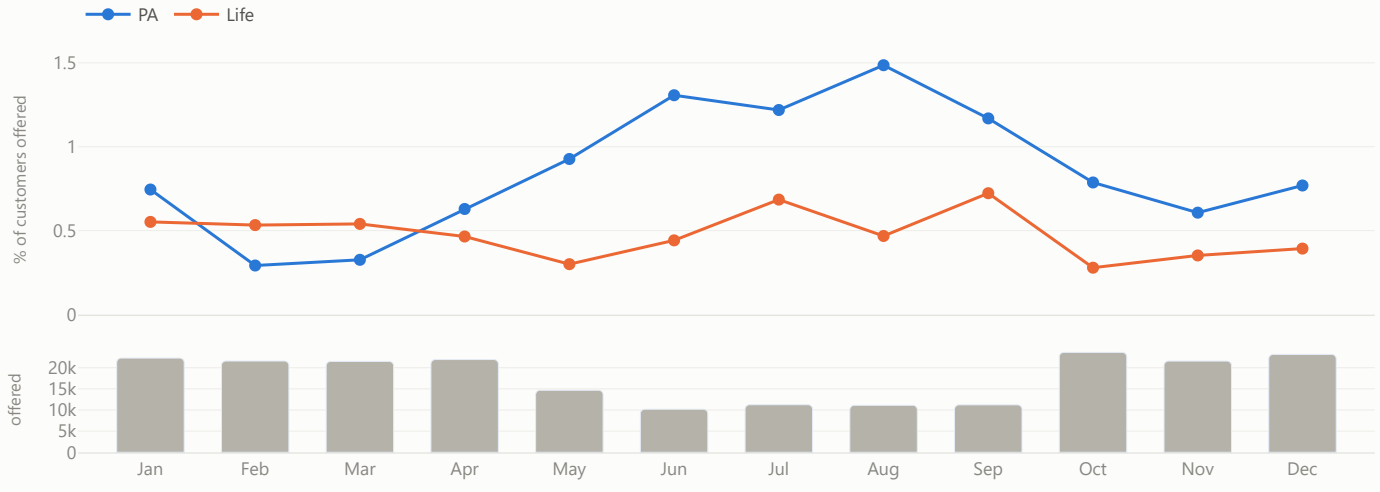
Under-40s earning 30k+ baht a month buy life at **0.96%**, 2.1× the life average. Dormant customers also lean to life, not PA.

1. Policy: who buys

Only 1.22% accept. PA outsells life about 3 to 2. Young customers buy both products more often; beyond that, PA and life attract different customers.

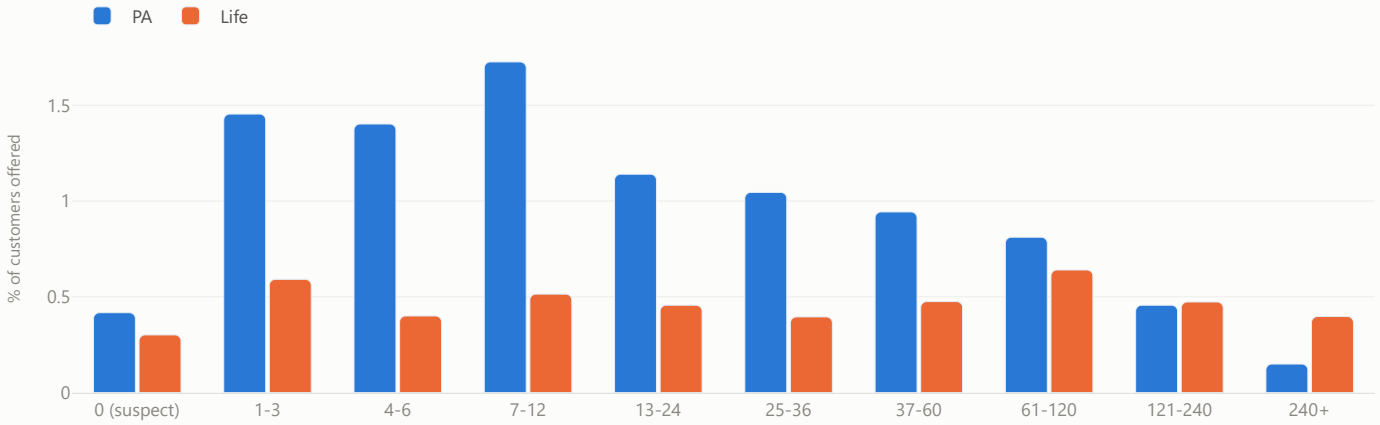
By campaign month

Top: acceptance rate. Bottom: number of customers offered. Jun–Sep had half the volume and the best rates. Oct–Feb also hold most of the suspect MOB-0 rows, so part of that gap is data, not season.



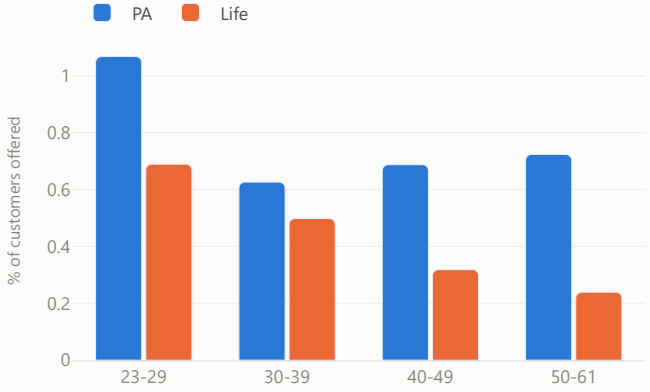
By months on book (MOB)

Customers who opened an account 1–12 months ago accept at **2.08%**, 1.7× the average, almost all from PA. PA falls steadily with tenure. MOB 0 (26% of customers) looks like missing data, not new customers: see Data quality.



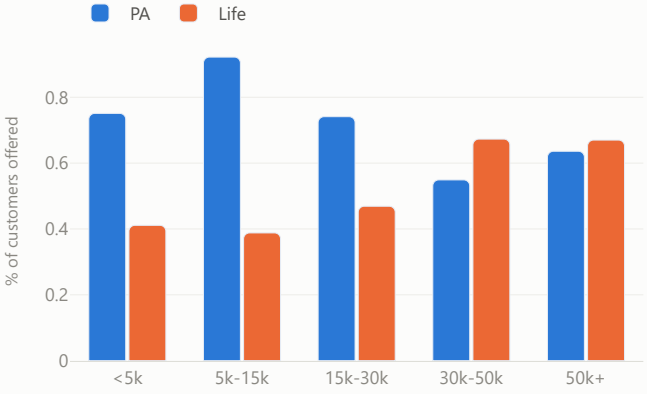
By age

Under-30s buy most. Life falls steadily with age; over-50s buy life at a third of the under-30 rate.



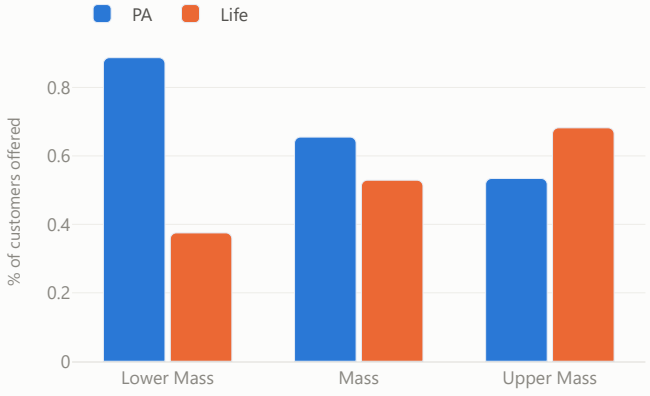
By monthly income (baht)

PA peaks at 5k–15k; life is highest from 30k up.



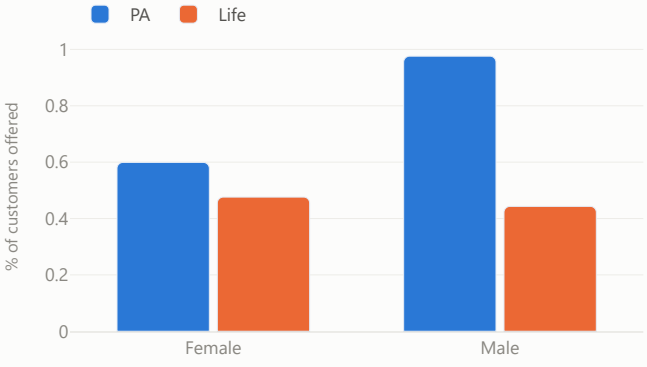
By segment

Lower Mass leans PA, Upper Mass leans life.



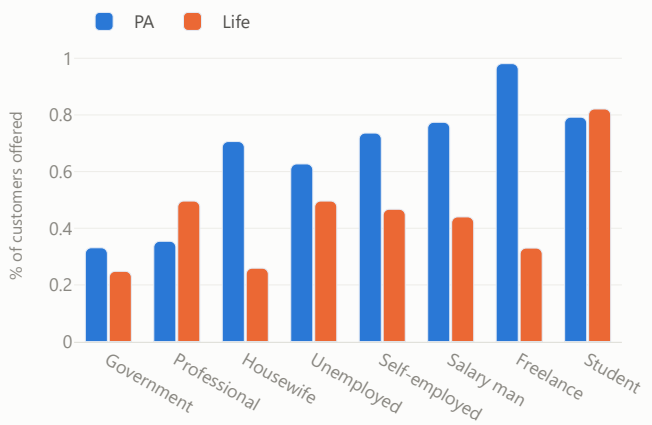
By gender

Men buy PA much more often; life is the same.



By occupation (≥ 1,000 customers)

Students buy life most; government staff buy least.

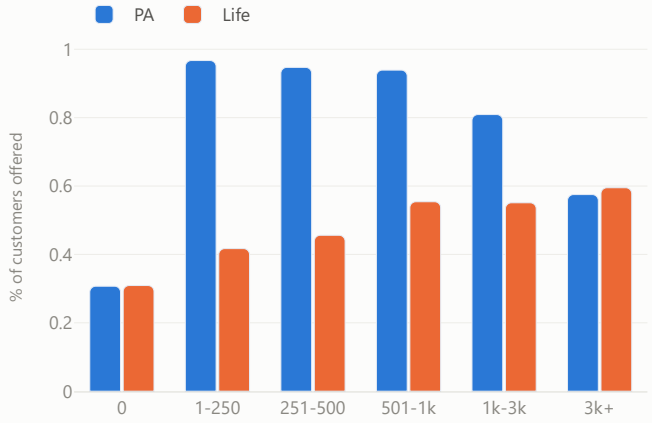


2. Payment and account activity

The clearest signal in the data. Customers who used their account in the last 30 days are far more likely to buy PA. For life, payment size matters more than activity.

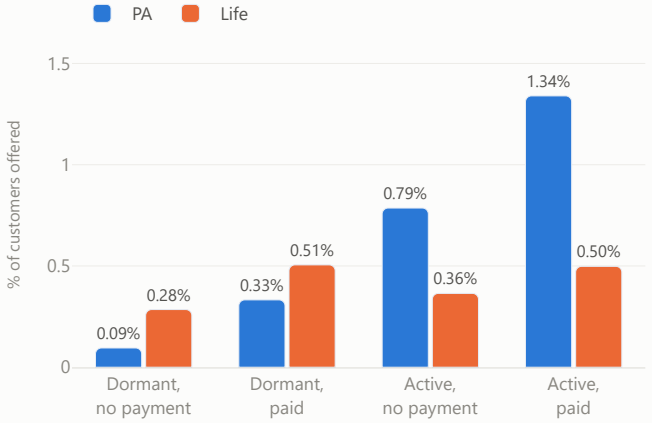
By payment in last 30 days (baht)

No payment → half the average rate. Any payment lifts PA most; life keeps rising with payment size.



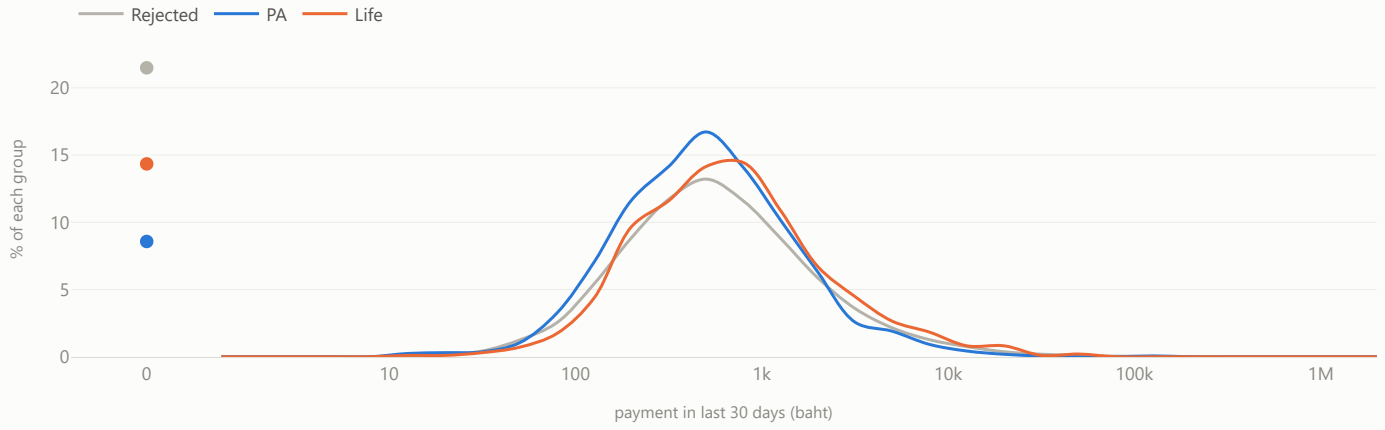
Account activity × payment

PA rate is 1.27% for active accounts vs 0.26% for dormant ones. Life barely changes.



How much each group paid

Distribution of 30-day payment. Dots at 0 = share who paid nothing: 21% of rejecters vs 9% of PA buyers and 14% of life buyers.

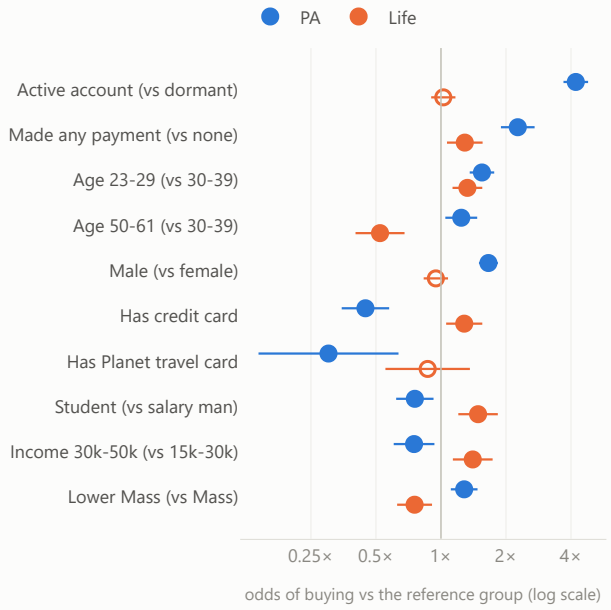


3. What matters once everything else is held constant

Logistic regression with all factors together, one model per product. Filled dot = significant at 5%; hollow = not significant. Right of 1× = more likely to buy.

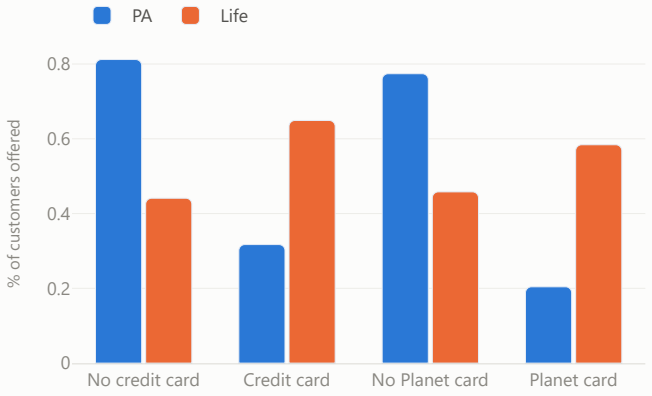
Odds of buying, by factor

Selected factors, with payment as yes/no. The full model with payment bands is in notebook 02.



Card holders swap PA for life

Credit-card and Planet travel-card holders rarely buy PA, possibly because the card already covers accidents or travel.



Simple targeting rule

−15% offers

Skipping customers who are **dormant and made no payment** cuts 15% of offers but loses only **1.8% of PA sales**. It would lose 9% of life sales, so send that group a life-only offer instead.

4. Combining the drivers: who to target

Crossing the strongest single factors (account activity with tenure for PA, age with income for life) gives segments that buy at 2–4× the average. Every pattern below shows up in both of two random halves of the data, so it is not a lucky cut.

PA: account activity × months on book

Each cell: PA rate, with the number of customers below it. Active accounts 1–12 months old reach 2.98%; dormant accounts stay near 0.1% at every tenure. MOB 0 with an active account: 0 PA buyers out of 2,732.

account activity, last 30 days

months on book	Dormant account	Active account
0 (suspect)	**0.44%** 53,412	**0.00%** 2,732
1-12	**0.07%** 4,593	**2.98%** 4,971
13-60	**0.08%** 25,954	**1.55%** 45,676
61-120	**0.09%** 14,635	**1.12%** 34,444
121+	**0.11%** 7,964	**0.57%** 18,372

Life: age × income

Life rate per cell. Under-40s earning 30k+ buy life at 0.96%; at every income, life falls with age.

monthly income (baht)

age	<5k	5k-15k	15k-30k	30k-50k	50k+
23-29	**0.69%** 18,291	**0.57%** 14,794	**0.63%** 13,982	**1.26%** 3,489	**0.92%** 1,515
30-39	**0.40%** 23,404	**0.39%** 18,060	**0.48%** 23,922	**0.83%** 7,247	**0.98%** 4,302
40-49	**0.28%** 16,867	**0.26%** 11,796	**0.37%** 15,215	**0.34%** 5,635	**0.42%** 3,781
50-61	**0.18%** 11,500	**0.18%** 6,630	**0.33%** 7,868	**0.24%** 2,950	**0.34%** 2,344

Targeting playbook

Segments can overlap; use them top-down. Rates are % of customers in the segment (averages: PA 0.76%, life 0.46%).

Segment	Customers	% of base	PA rate	Life rate	Share of PA / life buyers	Offer
Active, account opened 1–12 months ago	4,971	2.3%	**2.98%**	0.42%	9% / 2%	PA first, as part of onboarding
Active, account 13–120 months old	80,120	37.5%	**1.37%**	0.47%	67% / 38%	PA
Under 40, income 30k+ baht	16,824	7.9%	0.62%	**0.96%**	6% / 16%	Life
Dormant account (MOB 1+)	53,146	24.9%	0.09%	**0.61%**	3% / 33%	Life only; PA rarely sells here

5. Key takeaways

Findings

- **PA is an "active banker" product.** An active account gives about 4× the odds of buying PA, even with all other factors held constant.
- **Life is a "young, better-off" product.** Under-40s earning 30k+ buy life at 0.96%, about 2× the life average; students and bigger payers also lean to life.
- **New and active is the best PA segment.** Active accounts opened 1–12 months ago buy PA at 3.0%, 4× the PA average. Dormant accounts almost never buy PA but buy life above average (0.61%).
- **Card holders choose life over PA.**
- **Single factors are modest; combinations are not.** One factor alone moves rates about 2×; activity plus tenure reaches about 4× for PA.

Recommendations

- Follow the targeting playbook: PA at onboarding for new active accounts, PA for other active accounts, life for under-40 earners, life only for dormant accounts.
- Pitch life to under-30s and early-career earners; lead with life for card holders.
- Ask the campaign team why Jun–Sep converted better: was it different targeting, or a separate wave?
- Fix the data: check why MOB and account flows are 0 for a quarter of customers; add premium, policy status, campaign ID and year, and a customer ID.

6. Data quality

What was found and what was done. Full checks in notebook 01.

Issue	Rows	Decision
Exact duplicates, mostly null	1,940	Dropped: fewer than half of the 26 fields filled, so they add nothing beyond a rejection count.
Exact duplicates, fully populated	233	Kept: no customer ID to prove they are the same person. De-duplicate upstream, before IDs are masked.
15-day flow columns	100% of rows	Identical to the 30-day columns (extraction bug). Dropped.
currentacc_bal	100% of rows	Only takes 0 or 1, cannot be a balance. Dropped.
Missing financial fields	461	Kept for overall rates; skipped in payment/activity views.
Months on book longer than adult life	839	Set to missing for that field.
Months on book = 0	56,144 (26%)	Kept as is, but suspect. 96% have net flow of exactly 0 and 86% hold a balance with no inflow, which a new account can't do. Mostly in Oct–Feb. And the 2,732 MOB-0 customers with an active account bought no PA at all (about 35 expected). Likely unfilled fields: excluding them makes the PA activity effect stronger, not weaker.
Income = 0	342	Kept and flagged.
No premium or policy-status column	-	Revenue and premium collection cannot be measured.
No year or campaign ID	-	Month differences reported, not called seasonality.

Source: dsc_test_case.csv (215,993 raw rows → 214,053 after cleaning). Built by scripts/build_report.py from src/prep.py. Associations, not causes: the offer channel and message are unknown.